

CV Draft for Japanese Master's Applications

작성일: 2026-05-11

용도: 일본 석사 컨택/사전심사/출원용 CV 초안. 실제 제출 전에는 영문 1-2페이지 형식으로 정리한다.

0. Positioning Statement

Economics undergraduate with software/AI training, interested in applying machine learning, natural language processing, and agent-based simulation to financial and socio-economic systems.

한국어 설명:

경제학을 주전공으로 공부하며 경제·금융 시스템을 보는 도메인 감각을 쌓았고, 3학년 이후 소프트웨어/AI 쪽으로 연구 관심을 확장했다. 현재 포지션은 전통 경제학 연구자보다 financial and socio-economic systems를 다루는 AI/ML researcher or engineer 에 가깝다.

(상담용 : 왜 일본이어야 하는가?)

1. 상대적으로 괜찮은 석박사 연구환경
 2. 한국보다 조기에 정부주도하 인공지능 산업 투자 시작
 3. 다른 유학 후보(미국, 유럽 등)와 비교시 물리적 거리 및 문화적 거리가 비교적 가까움
- 각 후보별 탈락 이유
 - 미국 : 매우 높은 학업적 성취 기준, 박사 풀펀딩이 없다면 매우 높은 학업비&생활 물가
 - 유럽(독일) : 학업비는 미국보단 저렴. 하지만 마찬가지로 높은 학업적 기준, 문화적 이질성, 일정기준 미달시 커리큘럼 자격 자체 박탈(fail 3번로 기억), 마냥 좋진 않은 독일 취업시장
4. 국공립 대학의 경우 저렴한 등록금 및 장학금제도
 5. 일본에서의 석-박사간 크지 않은 임금격차
 6. 상대적으로 괜찮은 학사/석사 취업 시장 상황

1. Basic Information

항목	내용
Name	노동주 (盧東柱; No Dong-ju)
University	Inha University
Major	Economics
Additional major/minor	Interdisciplinary major in Software Convergence Engineering (連携専攻 : ソフトウェア融合工学 , from 2024-2)
Enrollment date	2021-03-02
Current status	Undergraduate student, 4th year
Cumulative GPA	4.18 / 4.50
Credits earned	136
Expected graduation	2027-March
Email	Ocean33@inha.edu
GitHub / Portfolio	https://github.com/Saturn294571
Languages	Korean native, English TOEFL target 80-90+, Japanese JLPT target N2/N1

2. Education

Undergraduate

- Major: Economics
- Additional academic focus: software, machine learning, data analysis
- Enrollment date: 2021-03-02
- Current cumulative GPA: 4.18/4.50
- Credits earned so far: 136
- Relevant coursework:
 - Economics: Microeconomics, Macroeconomics, History of Economics, The Analytic Study for Economic Problems, International Finance, Financial Economics, Economics of technology and innovation, Financial Investments, Global Economy, Economics of Social Security, Welfare Economics, Econometrics, Financial Econometrics
 - Quantitative methods: Mathematics for Economics, Statistics, Regression Analysis, Stastical Computing, Linear Algebra
 - Computing and data: Introduction to Computer Engineering, Data Structures, Algorithms, Computer Networks, Computer Systems, Database, Data Mining
 - Research training: Undergraduate Research Practice - AI General Research 3, Undergraduate Research Practice - AI General Research 4

3. Research Experience

LLM Economist++: History Filtering for LLM-based In-Context Reinforcement Learning in an Economic Simulation

Status: undergraduate research / paper draft
Keywords: LLM, in-context reinforcement learning, history filtering, economic simulation, agent-based modeling, computational efficiency

Summary:

- Implemented and evaluated a history selection mechanism inspired by Filtering Learning Histories within an LLM-based ICRL economic simulation framework.
- Built a simplified monopoly market environment with a Cobb-Douglas production function, demand function, and analytically computable optimal benchmark.
- Compared baseline ICRL and LHF-style filtered-history ICRL in terms of policy quality and computational cost.
- Found that LHF maintained near-optimal profit recovery while reducing prompt length and execution cost.

Key result draft:

Metric	Baseline	LHF
Profit mean	68.9485	68.6464
Profit std	0.5621	0.4904
Recovery ratio	0.9955	0.9911

Limitations to state honestly:

- Single or limited seed setting
- Simplified static monopoly environment
- Simple top-k history filtering rule
- Needs robustness checks across dynamic, multi-agent, and parameter-varied environments

4. Project Experience

Selected Projects

Project Cassandra: 2025 Economics Academic Festival

Topic: "Why did the exchange rate surge despite strong exports in 2025?"
Format: economics research project / festival paper / award-winning presentation

- Analyzed the 2025 Korean macroeconomic "export-exchange rate paradox" using a modified IS-LM-BP framework with OLS, VAR, and impulse response analysis.
- Built and used an AI-assisted research workflow for research design, data collection support, coding, visualization, and paper drafting.
- Produced a full paper and empirical figures based on monthly macroeconomic data (2015-2025), focusing on fiscal dominance and risk premium channels.
- Received an encouragement prize (奨励賞) in an on-campus AI-themed economics academic festival.

Festival Compass in Seoul: ML-based Recommendation Service

Topic: recommendation system for reducing tourism concentration in Seoul

Format: club ML service contest project

- Developed the core recommendation model using TF-IDF and cosine similarity for culture/festival recommendation.
- Built and deployed a FastAPI backend service, connecting offline model artifacts with online API inference.
- Worked on backend and ML integration with Python, pandas, scikit-learn, FastAPI, SQLite, Docker, and Google Cloud Run.
- Designed a two-track recommendation logic combining similarity-based retrieval and less-visited district recommendation.

Economic / Financial AI Project Candidates

These are not yet all completed outputs. Use only completed work in the final CV.

- Financial/economic sentiment analysis
- Financial time-series forecasting with transformer-based models
- Credit forecasting
- Graph-based modeling of asset markets
- LLM/agent-based market simulation

Previous Activities

- On-campus AI-themed economics academic festival: encouragement prize (奨励賞), research paper and presentation completed
- Club ML service contest: recommendation service prototype built and deployed
- Undergraduate research assistant activity: detail TBD

5. Technical Skills

Draft categories:

- Programming: Python, TBD
- Data analysis: pandas, NumPy, statsmodels, scikit-learn, TBD
- Machine learning: supervised learning, time-series modeling, NLP, LLM API use, TBD
- Research tools: LaTeX/Markdown, Git/GitHub, Zotero or reference manager TBD
- Econometrics/statistics: regression, time-series, hypothesis testing, financial econometrics, TBD

6. Research Interests

Primary:

- Financial AI
- Socio-economic AI
- Computational economics
- LLM/agent-based simulation
- NLP for economic and social text data

Possible proposal titles:

- Interpretable Financial AI for Text-Market Data Fusion and Risk Forecasting
- LLM/Agent-based Simulation for Socio-economic Decision Making and Market Behavior
- Robust History Selection and Stabilization in LLM-based Economic Simulation

7. Admission Target Fit

Strong fit:

- NAIST Information Science: NLP, social computing, AI-oriented research
- Tsukuba Risk and Resilience Engineering: risk, resilience, engineering methods, social/financial systems
- Kobe KIMAP/Economics: financial AI and economics background
- Kyushu Economics: computational economics and multi-agent simulation
- Hitotsubashi/Shiga/Kyoto: social data science and social informatics

8. Missing Information Checklist

- TOEFL/JLPT scores or expected test dates
- Paper title/status
- Supervisor/lab name
- Exact contest/festival official names and dates
- Awards, presentations, scholarships
- Concrete technical stack used in LLM Economist++